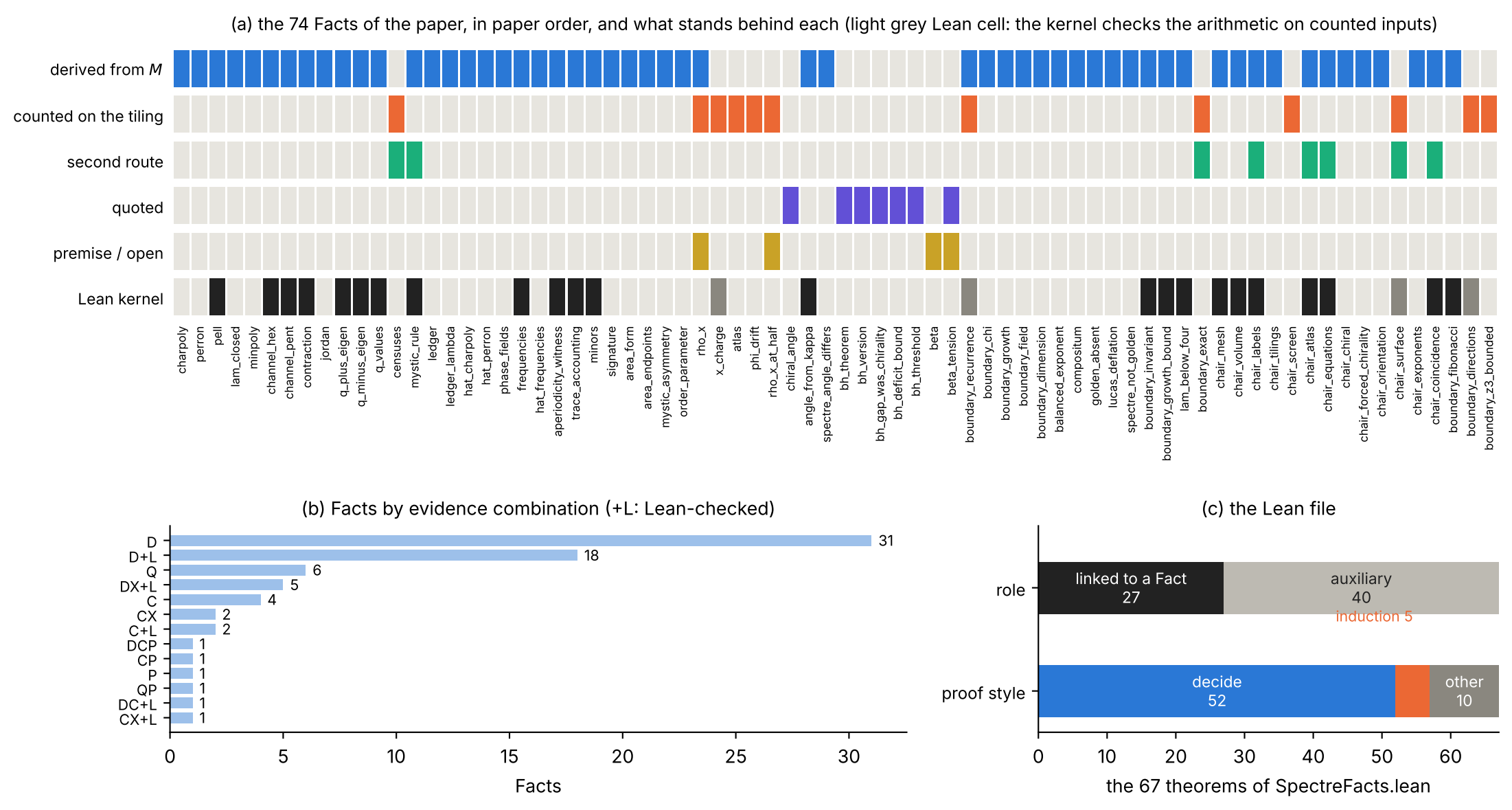
(a) the 74 Facts of the paper, in paper order, and what stands behind each (light grey Lean cell: the kernel checks the arithmetic on counted inputs)
derived from M
counted on the tiling
second route
quoted
premise / open
Lean kernel
charpoly
perron
pell
lam_closed
minpoly
channel_hex
channel_pent
contraction
jordan
q_plus_eigen
q_minus_eigen
q_values
censuses
mystic_rule
ledger
ledger_lambda
hat_charpoly
hat_perron
phase_fields
frequencies
hat_frequencies
aperiodicity_witness
trace_accounting
minors
signature
area_form
area_endpoints
mystic_asymmetry
order_parameter
rho_x
x_charge
atlas
phi_drift
rho_x_at_half
chiral_angle
angle_from_kappa
spectre_angle_differs
bh_theorem
bh_version
bh_gap_was_chirality
bh_deficit_bound
bh_threshold
beta
beta_tension
boundary_recurrence
boundary_chi
boundary_growth
boundary_field
boundary_dimension
balanced_exponent
compositum
golden_absent
lucas_deflation
spectre_not_golden
boundary_invariant
boundary_growth_bound
lam_below_four
boundary_exact
chair_mesh
chair_volume
chair_labels
chair_tilings
chair_screen
chair_atlas
chair_equations
chair_chiral
chair_forced_chirality
chair_orientation
chair_surface
chair_exponents
chair_coincidence
boundary_fibonacci
boundary_directions
boundary_z3_bounded
(b) Facts by evidence combination (+L: Lean-checked)
D 31
D+L 18
Q 6
DX+L 5
C 4
CX 2
C+L 2
DCP 1
CP 1
P 1
QP 1
DC+L 1
CX+L 1
0
5
10
15
20
25
30
Facts
(c) the Lean file
role
linked to a Fact 27
auxiliary 40
induction 5
proof style
decide 52
other 10
0
10
20
30
40
50
60
the 67 theorems of SpectreFacts.lean